
Data Mining:

Concepts and Techniques

(3rd ed.)

— Chapter 12 —

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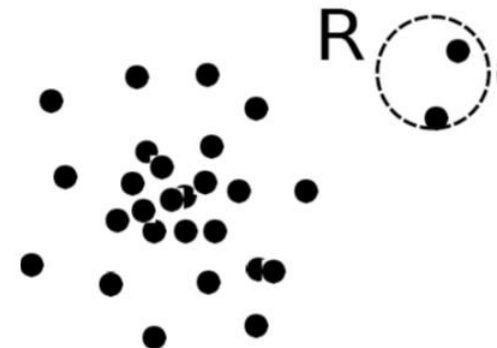
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Chapter 12. Outlier Analysis

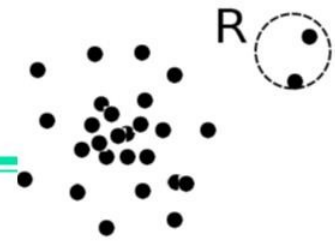
- Outlier and Outlier Analysis 
- Outlier Detection Methods
- Statistical Approaches
- Proximity-Base Approaches
- Clustering-Base Approaches
- Classification Approaches
- Mining Contextual and Collective Outliers
- Outlier Detection in High Dimensional Data
- Summary

What Are Outliers?

- **Outlier:** A data object that **deviates significantly** from the normal objects as if it were **generated by a different mechanism**
 - Ex.: Unusual credit card purchase, sports: Michael Jordon, Wayne Gretzky, ...
- Outliers are different from the noise data
 - Noise is random error or variance in a measured variable
 - Noise should be removed before outlier detection
- Outliers are interesting: It violates the mechanism that generates the normal data
- Outlier detection vs. *novelty detection*: early stage, outlier; but later merged into the model
- Applications:
 - Credit card fraud detection
 - Telecom fraud detection
 - Customer segmentation
 - Medical analysis



Types of Outliers (I)



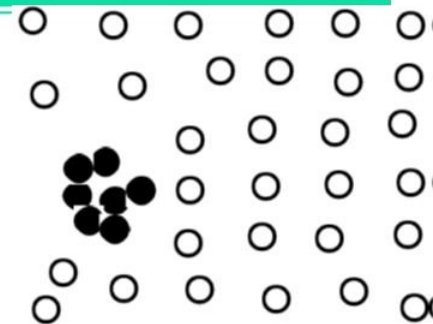
Global Outlier

- Three kinds: *global*, *contextual* and *collective* outliers
- **Global outlier** (or point anomaly)
 - Object is O_g if it significantly deviates from the rest of the data set
 - Ex. Intrusion detection in computer networks
 - Issue: Find an appropriate measurement of deviation
- **Contextual outlier** (or *conditional outlier*)
 - Object is O_c if it deviates significantly based on a selected context
 - Ex. 80° F in Urbana: outlier? (depending on summer or winter?)
 - Attributes of data objects should be divided into two groups
 - Contextual attributes: defines the context, e.g., time & location
 - Behavioral attributes: characteristics of the object, used in outlier evaluation, e.g., temperature
 - Can be viewed as a generalization of *local outliers*—whose density significantly deviates from its local area
 - Issue: How to define or formulate meaningful context?

Types of Outliers (II)

■ Collective Outliers

- A subset of data objects *collectively* deviate significantly from the whole data set, even if the individual data objects may not be outliers
- Applications: E.g., *intrusion detection*:
 - When a number of computers keep sending denial-of-service packages to each other
- Detection of collective outliers
 - Consider not only behavior of individual objects, but also that of groups of objects
 - Need to have the background knowledge on the relationship among data objects, such as a distance or similarity measure on objects.
- A data set may have multiple types of outlier
- One object may belong to more than one type of outlier



Collective Outlier

Challenges of Outlier Detection

- Modeling normal objects and outliers properly
 - Hard to enumerate all possible normal behaviors in an application
 - The border between normal and outlier objects is often a gray area
- Application-specific outlier detection
 - Choice of distance measure among objects and the model of relationship among objects are often application-dependent
 - E.g., clinic data: a small deviation could be an outlier; while in marketing analysis, larger fluctuations
- Handling noise in outlier detection
 - Noise may distort the normal objects and blur the distinction between normal objects and outliers. It may help hide outliers and reduce the effectiveness of outlier detection
- Understandability
 - Understand why these are outliers: Justification of the detection
 - Specify the degree of an outlier: the unlikelihood of the object being generated by a normal mechanism

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Outlier Detection I: Supervised Methods

- Two ways to categorize outlier detection methods:
 - Based on whether user-labeled examples of outliers can be obtained:
 - Supervised, semi-supervised vs. unsupervised methods
 - Based on assumptions about normal data and outliers:
 - Statistical, proximity-based, and clustering-based methods
- **Outlier Detection I: Supervised Methods**
 - Modeling outlier detection as a classification problem
 - Samples examined by domain experts used for training & testing
 - Methods for Learning a classifier for outlier detection effectively:
 - Model normal objects & report those not matching the model as outliers, or
 - Model outliers and treat those not matching the model as normal
 - Challenges
 - Imbalanced classes, i.e., outliers are rare: Boost the outlier class and make up some artificial outliers
 - Catch as many outliers as possible, i.e., recall is more important than accuracy (i.e., not mislabeling normal objects as outliers)

Outlier Detection II: Unsupervised Methods

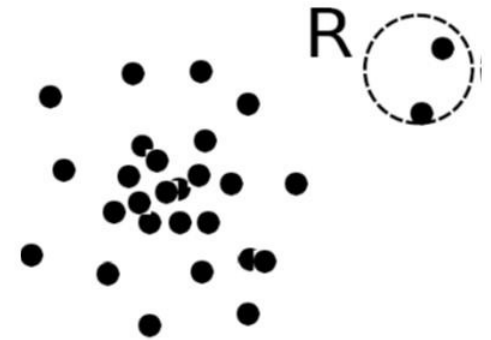
- Assume the normal objects are somewhat “clustered” into multiple groups, each having some distinct features
- An outlier is expected to be far away from any groups of normal objects
- Weakness: Cannot detect collective outlier effectively
 - Normal objects may not share any strong patterns, but the collective outliers may share high similarity in a small area
- Ex. In some intrusion or virus detection, normal activities are diverse
 - Unsupervised methods may have a high false positive rate but still miss many real outliers.
 - Supervised methods can be more effective, e.g., identify attacking some key resources
- Many clustering methods can be adapted for unsupervised methods
 - Find clusters, then outliers: not belonging to any cluster
 - Problem 1: Hard to distinguish noise from outliers
 - Problem 2: Costly since first clustering: but far less outliers than normal objects
 - Newer methods: tackle outliers directly

Outlier Detection III: Semi-Supervised Methods

- Situation: In many applications, the number of labeled data is often small: Labels could be on outliers only, normal objects only, or both
- Semi-supervised outlier detection: Regarded as applications of semi-supervised learning
- If some labeled normal objects are available
 - Use the labeled examples and the proximate unlabeled objects to train a model for normal objects
 - Those not fitting the model of normal objects are detected as outliers
- If only some labeled outliers are available, a small number of labeled outliers may not cover the possible outliers well
 - To improve the quality of outlier detection, one can get help from models for normal objects learned from unsupervised methods

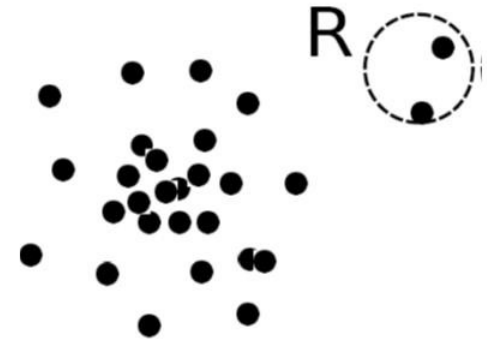
Outlier Detection (2): Proximity-Based Methods

- An object is an outlier if the nearest neighbors of the object are far away, i.e., the **proximity** of the object is **significantly deviates** from the proximity of most of the other objects in the same data set
- Example (right figure): Model the proximity of an object using its 3 nearest neighbors
 - Objects in region R are substantially different from other objects in the data set.
 - Thus the objects in R are outliers
- The effectiveness of proximity-based methods highly relies on the proximity measure.
- In some applications, proximity or distance measures cannot be obtained easily.
- Often have a difficulty in finding a group of outliers which stay close to each other
- Two major types of proximity-based outlier detection
 - Distance-based vs. density-based



Outlier Detection (3): Clustering-Based Methods

- Normal data belong to large and dense clusters, whereas outliers belong to small or sparse clusters, or do not belong to any clusters
- Example (right figure): two clusters
 - All points not in R form a large cluster
 - The two points in R form a tiny cluster, thus are outliers
- Since there are many clustering methods, there are many clustering-based outlier detection methods as well
- Clustering is expensive: straightforward adaption of a clustering method for outlier detection can be costly and does not scale up well for large data sets



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Proximity-Based Approaches: Distance-Based vs. Density-Based Outlier Detection

- Intuition: Objects that are far away from the others are outliers
- Assumption of proximity-based approach: The proximity of an outlier deviates significantly from that of most of the others in the data set
- Two types of proximity-based outlier detection methods
 - Distance-based outlier detection: An object o is an outlier if its neighborhood does not have enough other points
 - Density-based outlier detection: An object o is an outlier if its density is relatively much lower than that of its neighbors

Distance-Based Outlier Detection

- For each object o , examine the # of other objects in the r -neighborhood of o , where r is a user-specified **distance threshold**
- An object o is an outlier if most (taking π as a **fraction threshold**) of the objects in D are far away from o , i.e., not in the r -neighborhood of o
- An object o is a $DB(r, \pi)$ outlier if
$$\frac{||\{o' | dist(o, o') \leq r\}||}{||D||} \leq \pi$$
- Equivalently, one can check the distance between o and its k -th nearest neighbor o_k , where $k = \lceil \pi ||D|| \rceil$. o is an outlier if $dist(o, o_k) > r$
- Efficient computation: Nested loop algorithm
 - For any object o_i , calculate its distance from other objects, and count the # of other objects in the r -neighborhood.
 - If $\pi \cdot n$ other objects are within r distance, terminate the inner loop
 - Otherwise, o_i is a $DB(r, \pi)$ outlier
- Efficiency: Actually CPU time is not $O(n^2)$ but linear to the data set size since for most non-outlier objects, the inner loop terminates early

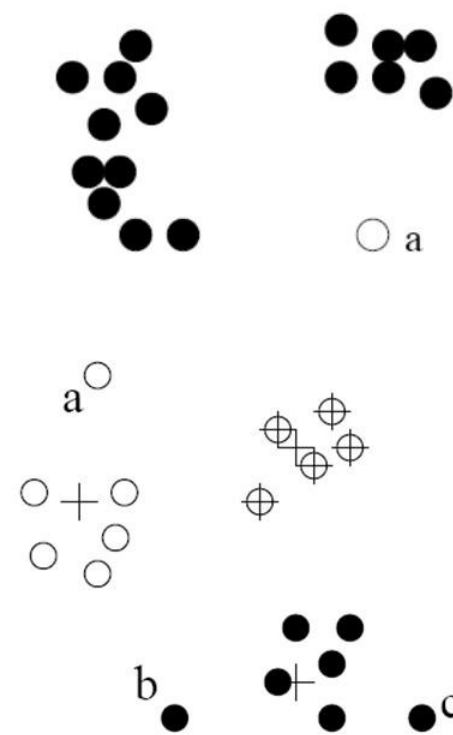
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Clustering-Based Outlier Detection (1 & 2):

Not belong to any cluster, or far from the closest one

- An object is an outlier if (1) it does not belong to any cluster, (2) there is a large distance between the object and its closest cluster, or (3) it belongs to a small or sparse cluster
- Case 1: Not belong to any cluster
 - Identify animals not part of a flock: Using a density-based clustering method such as DBSCAN
- Case 2: Far from its closest cluster
 - Using k-means, partition data points of into clusters
 - For each object o , assign an outlier score based on its distance from its closest center
 - If $\text{dist}(o, c_o) / \text{avg_dist}(c_o)$ is large, likely an outlier
- Ex. Intrusion detection: Consider the similarity between data points and the clusters in a training data set
 - Use a training set to find patterns of “normal” data, e.g., frequent itemsets in each segment, and cluster similar connections into groups
 - Compare new data points with the clusters mined—Outliers are possible attacks



Clustering-Based Method: Strength and Weakness

■ Strength

- Detect outliers without requiring any labeled data
- Work for many types of data
- Clusters can be regarded as summaries of the data
- Once the cluster are obtained, need only compare any object against the clusters to determine whether it is an outlier (fast)

■ Weakness

- Effectiveness depends highly on the clustering method used—they may not be optimized for outlier detection
- High computational cost: Need to first find clusters
- A method to reduce the cost: Fixed-width clustering
 - A point is assigned to a cluster if the center of the cluster is within a pre-defined distance threshold from the point
 - If a point cannot be assigned to any existing cluster, a new cluster is created and the distance threshold may be learned from the training data under certain conditions